

VIVEKA-INSIGHT: a cross-lingual concept graph and citation-grounded retrieval resource over the complete works of Swami Vivekananda in English and Bengali

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Abstract

Classical spiritual and philosophical corpora pose three compounding challenges for language resources: they exist in several languages without parallel alignment, their vocabulary differs sharply from that of contemporary readers, and any generated text over such culturally sensitive material must be verifiably grounded in the source. We present VIVEKA-INSIGHT, a bilingual resource and accompanying open-source pipeline for the works of Swami Vivekananda (1863–1902) — the nine-volume English *Complete Works* and the ten-volume Bengali *Vani o Rachana*, two related but non-parallel corpora totalling about 15 million characters. The released resource has four layers: (i) a structure-preserving parse of both corpora into a volume → chapter → paragraph → sentence hierarchy (32,694 paragraphs, 168,842 sentences) with per-paragraph anchors that deep-link back to the published editions; (ii) a *cross-lingual concept graph* of 8,362 language-agnostic concepts carrying 87,518 paragraph–concept edges typed by relation and 55,872 concept–concept edges, in which canonical English labels act as a string-equality key that links Bengali and English passages with no parallel data; (iii) a bilingual alias inventory of 60,850 surface forms (30,053 English, 30,797 Bengali) mapping each concept to its realizations in both languages; and (iv) a human-annotated evaluation set of 200 paragraph–concept edges judged independently by three annotators, released with all per-annotator judgments. We document the construction pipeline, which needs one extraction pass by a mid-sized instruction-tuned LLM and rebuilds incrementally in 10–15 minutes after source edits, and report three evaluations: known-item cross-lingual retrieval over

194 automatically verified rendered lecture pairs (Recall@10 0.86 in both directions); a 30-question audit of citation integrity and modern-question bridging; and the human study, which places concept-extraction precision at 0.60 under strict two-annotator consensus (Cohen’s $\kappa = \mathbf{0.61}$). The study also shows the extractor’s confidence weight is calibrated — restricting to weight $\geq \mathbf{0.8}$ raises precision to 0.71 while retaining 98% of concept-bearing paragraphs — and that precision is markedly lower in Bengali than English (0.54 vs 0.68), locating the resource’s weakness in exactly the half that cross-lingual access depends on. The design assumes nothing Vivekananda-specific and transfers to other multilingual classical corpora.

Keywords: cross-lingual concept graph, non-parallel bilingual corpus, Bengali language resources, retrieval-augmented generation, digital humanities, cultural heritage corpora

1 Introduction

Large digitized editions of classical religious and philosophical literature are now widely available, but access to them remains largely lexical: readers search for words, while what they hold are questions. The gap is widest for corpora like the works of Swami Vivekananda, a central figure of modern Vedanta whose collected English works span nine volumes and whose Bengali writings and recorded conversations fill ten more. The two collections overlap thematically but are *not* translations of one another; a reader fluent in only one language has no lexical route into roughly half of the material.

A second, subtler gap is *temporal*. Contemporary readers bring contemporary questions — about smartphone addiction, social-media envy, or career anxiety — to an author who died in 1902. The vocabulary of such questions simply does not occur in the corpus, so even strong dense retrievers underperform; yet the corpus deals extensively with the underlying phenomena under names such as *attachment*, *control of the mind*, and *habit*. Bridging this gap is not a retrieval nicety but the central requirement for making a nineteenth-century corpus useful to twenty-first-century readers.

Finally, any system that *generates* answers about revered texts must confront hallucination [13]. Attributing invented statements to a religious figure is a materially worse failure mode than a wrong answer in open-domain QA, so verifiable grounding — every claim traceable to a specific passage a reader can open — is a hard constraint rather than a desirable extra.

This paper contributes a *resource* built around these three requirements, together with the open pipeline that produces it and the evaluations that characterize its quality.

The resource.

Four layers are released (Section 5 documents formats and distribution):

1. **A structured bilingual parse.** Both published editions are recovered into a volume $\rightarrow$ chapter $\rightarrow$ paragraph $\rightarrow$ sentence hierarchy — 1,991 chapters, 32,694

paragraphs, 168,842 sentences — with the HTML anchor of every paragraph retained, so any unit in the resource deep-links to its exact location in the public online editions (Section 3).

2. **A cross-lingual concept graph.** An instruction-tuned LLM reads every paragraph of both corpora and emits concepts with *canonical English, lowercase, hyphenated labels*. Because the label space is shared, a Bengali paragraph about *vairagya* and an English lecture on renunciation attach to the same node; string equality on labels yields cross-lingual linking without any parallel data. The graph carries 8,362 concepts, 87,518 relation-typed paragraph–concept edges, and 55,872 concept–concept edges (Section 4).
3. **A bilingual alias inventory.** 60,850 surface forms (30,053 English, 30,797 Bengali) record how each concept is actually realized in each language — usable independently of the rest of the resource as a concept-level bilingual lexicon for Vedantic vocabulary (Section 4).
4. **A human-annotated evaluation set.** 200 paragraph–concept edges sampled uniformly at random, each judged independently by three annotators, released with all per-annotator judgments so that alternative aggregations and agreement analyses can be recomputed (Section 9.3).

The pipeline and its evaluation.

Around the resource we document and release: multi-granular fused retrieval, which matches queries against sentence-, paragraph- and chapter-level indexes in both languages *and* against the concept graph, fusing candidates with weighted reciprocal rank fusion [8] and reranking with a multilingual cross-encoder (Section 6); temporal concept bridging, which maps modern questions onto the concept vocabulary that verifiably exists in the graph and surfaces the mapping as an inspectable “bridge” (Section 7); citation-grounded generation, in which every claim carries a citation rewritten into a deep link to the exact source paragraph and unsupported markers are deleted (Section 7); and an economical build — one GPU-day for the full corpus, 10–15 minutes for an incremental rebuild after a source edit, which is what makes the resource maintainable rather than frozen (Section 4.2).

2 Related Work

Dense retrieval and RAG. Dense passage retrieval [2] and retrieval-augmented generation [1, 3] are the standard architecture for grounded QA. We adopt the general recipe but depart in two ways: the retrieval layer is multi-granular and graph-augmented, and generation is constrained to a citation discipline enforced by post-processing rather than trust in the model.

Graph-augmented retrieval. GraphRAG [4] builds an entity graph over a corpus with an LLM and uses community summaries for query-focused summarization. VIVEKA-INSIGHT shares the LLM-builds-a-graph idea but uses the graph differently: nodes are *concepts* with language-agnostic canonical labels, and the graph’s primary role is cross-lingual bridging and query-time expansion into two non-parallel corpora, rather than corpus summarization.

Multilingual embeddings. We build on BGE-M3 [5], whose single embedding space covers both English and Bengali and supports inputs long enough for chapter-level representation, and on its companion multilingual cross-encoder reranker. Sentence-level transformer embeddings follow the general approach of [6].

QA over religious corpora. Verse-based test collections and QA systems exist for the Qur’an [17] and other scriptures; these typically operate in a single language over a closed canon. To our knowledge, VIVEKA-INSIGHT is the first system to combine non-parallel bilingual retrieval, an LLM-built concept layer, and explicit modern-question bridging for a classical corpus.

Positioning as a resource. Existing cultural-heritage language resources of this kind fall into three groups, and this resource sits between them. Digitized *editions* (page images, TEI transcriptions) preserve structure and provenance but offer no semantic access layer. *Aligned bilingual corpora* support cross-lingual work but presuppose translation pairs, which do not exist here and cannot be manufactured without distorting the material: the Bengali collection is not a translation of the English one. *Knowledge graphs* over cultural heritage are typically entity-centric — people, places, works — and built by linking to external authority files, which for a body of philosophical prose captures the proper nouns and misses the argument. The layer released here is instead *concept*-centric, built from the text rather than from an authority file, and language-agnostic by construction, which is what allows non-parallel corpora to be linked at all. For Bengali specifically, where resources remain comparatively scarce, the alias inventory and the annotated edge sample also constitute reusable artefacts in their own right.

3 Corpus and Preprocessing

The English source is the nine-volume *Complete Works of Swami Vivekananda* (lectures, letters, recorded conversations, and writings; ~8.1 million characters). The Bengali source is the ten-volume *Vani o Rachana* (~6.9 million characters), which contains original Bengali writings and conversations as well as independent Bengali renderings of some English lectures; no paragraph-level alignment between the two exists, and we do not attempt to create one.

Both corpora are parsed from their published HTML editions with structure-aware parsers that recover the volume → (section) → chapter → paragraph hierarchy and preserve both the chapter and the per-paragraph HTML anchor identifiers, which later enable paragraph-precise deep-linking of citations back into the publicly hosted editions. Sentences are split with language-appropriate rules (NLTK [12] for English; a rule-based splitter honouring the *danda* and Bengali punctuation for Bengali). Bengali text is NFC-normalized throughout; a single normalization function is the sole source of truth for text keys across the pipeline, which the incremental-rebuild machinery (Section 4.2) depends on. Table 1 summarizes the corpus.

Table 1 Corpus statistics after parsing.

	English	Bengali	Total
Volumes (published)	9	10	19
Chapters	1,452	539	1,991
Paragraphs	18,239	14,455	32,694
Sentences	88,150	80,692	168,842
Characters	8.1M	6.9M	15.0M

Table 2 Concept-graph statistics.

Concepts (canonical labels)	8,362
Concept aliases (EN / BN)	30,053 / 30,797
Entities (person / other / place / text / deity)	3,688 / 2,372 / 1,275 / 846 / 492
Paragraph→concept edges	87,518
discusses / exemplifies / contrasts	75,403 / 8,735 / 3,380
Concept↔concept edges	55,872
similar / co-occurs	29,482 / 26,390
Paragraphs with ≥ 1 concept	23,363 (71.5%)
retained at extractor weight ≥ 0.8	22,914 (98.1%)

4 The Concept Graph

4.1 LLM concept extraction

An instruction-tuned LLM (Qwen2.5-14B-Instruct [9], served with vLLM [10]) reads every paragraph in both languages and emits structured JSON containing (i) a one-line English summary, (ii) a set of *concepts*, and (iii) a set of named *entities* typed as person, place, text, or deity. The extraction prompt imposes the single most important invariant of the system: **concept labels are canonical English, lowercase, and hyphenated** regardless of the paragraph language. The canonical label thus acts as a cross-lingual key: the Bengali surface form and the English surface form of a concept are stored as *aliases* (60,838 aliases; 30,051 English, 30,787 Bengali), while identity lives in the label itself. Each paragraph–concept edge carries a relation (*discusses*, *exemplifies*, or *contrasts*) and a weight.

Near-duplicate concepts produced by paraphrase (e.g., spelling variants) are merged by embedding the labels and clustering above a conservative cosine threshold (0.92). Two kinds of concept–concept edges are then built: *similar* edges (label-embedding cosine ≥ 0.78 , top-12 neighbours per node) and *co-occurs* edges (concepts extracted from the same paragraph at least twice corpus-wide). Table 2 summarizes the resulting graph; Table 3 lists the most frequent concepts, which align well with the corpus’s known thematic centres.

The 71.5% paragraph coverage is deliberate rather than a deficiency: short fragments (single-word replies in recorded conversations, headings, brief exclamations) legitimately carry no concept content, and forcing extraction on them degrades graph precision.

Table 3 Most frequent concepts by paragraph mentions.

Concept	Mentions	Concept	Mentions
self-realization	4,521	renunciation	2,553
knowledge-realization	3,858	liberation	2,528
devotion	3,179	patience	1,787
maya	2,684	duty	1,437
non-attachment	2,669	karma	1,334

Coverage, however, is not correctness. A two-annotator study of the edges themselves (Section 9.3) puts their strict-consensus precision at 0.60, rising to 0.71 on the high-confidence subgraph and falling to 0.54 on Bengali material; the graph should be read as a broad, imperfect expansion layer rather than a curated index, and everything we claim for it below is calibrated to that.

4.2 Economical construction and incremental rebuilds

Concept extraction is the only expensive stage (~ 3.5 hours for the full corpus on a single 80 GB-class GPU); everything else (parsing, embedding, graph assembly) takes minutes. To make the corpus *editable* — OCR fixes and structural cleanups are routine in digitized classical texts — the pipeline snapshots all extraction outputs keyed by (NFC-normalized paragraph text, language, extractor version) before any re-parse, and restores them onto the fresh paragraph rows afterwards. Only genuinely new or changed paragraphs reach the LLM. An incremental rebuild after a content edit completes in 10–15 minutes, which in practice is the difference between a maintained resource and a frozen one.

5 The Released Resource

5.1 Data model and distribution format

The resource is distributed as a single normalized SQLite database together with the dense indexes built from it. SQLite was chosen deliberately over a graph database or a bespoke format: it needs no server, is readable by standard tooling in every major language, travels as one file, and keeps the corpus, the concept layer, and their provenance in one queryable place. Table 4 lists the released layers and their sizes.

Every unit carries its provenance. A paragraph row records its chapter, its ordinal position, its language, and the HTML anchor (`para_id.html`) of the paragraph in the published online edition, so any row in any layer can be resolved to a URL that opens the exact passage a reader needs to check. Each paragraph–concept edge carries the relation the extractor assigned (*discusses*, *exemplifies*, *contrasts*) and its confidence weight, which Section 9.3 shows to be calibrated and therefore usable as a quality filter by downstream reusers.

Table 4 Layers of the released resource. All relational layers live in a single 232 MB SQLite file; the dense indexes add 868 MB.

Layer	Relation	Units
Structural parse	books	2
	volumes	10
	chapters	1,991
	paragraphs	32,694
	sentences	168,842
Concept layer	concepts	8,362
	concept_aliases	60,850
	para_concept	87,518
	concept_edges	55,872
Entity layer	entities	8,673
	para_entity	41,937
Dense indexes	7 FAISS indexes	211,889 vectors
Human annotations	200 items $\times$ 3 annotators	600 judgments

5.2 What each layer offers independently

The layers are useful separately, not only as inputs to the retrieval stack described below:

- The **structural parse** is, to our knowledge, the first machine-readable sentence-segmented edition of either corpus with stable per-paragraph identifiers, and is directly reusable for stylometry, citation studies, or translation research.
- The **concept graph** supplies cross-lingual links between two corpora for which no parallel data exists, and can be used as distant supervision for training bilingual retrievers.
- The **alias inventory** — 30,053 English and 30,797 Bengali surface forms grouped under 8,362 canonical concepts — is usable on its own as a concept-level bilingual lexicon of Vedantic vocabulary, a domain poorly covered by general-purpose bilingual dictionaries.
- The **annotation set** ships with every individual annotator’s judgments rather than an aggregated gold label, so that reusers can apply their own aggregation, recompute agreement, or study annotator disagreement on semantic-relatedness judgments in a bilingual setting.

5.3 Availability and licensing

The pipeline source code is released under the MIT licence at <https://github.com/tamalrkm/viveka-insight>. The derived data layers — the structural parse, the concept graph, the alias inventory, the entity layer, and the annotation set — are released under CC BY 4.0 and archived with a persistent identifier at [10.5281/zenodo.21669974](https://doi.org/10.5281/zenodo.21669974). The dense indexes are rebuildable from the database and the released pipeline in

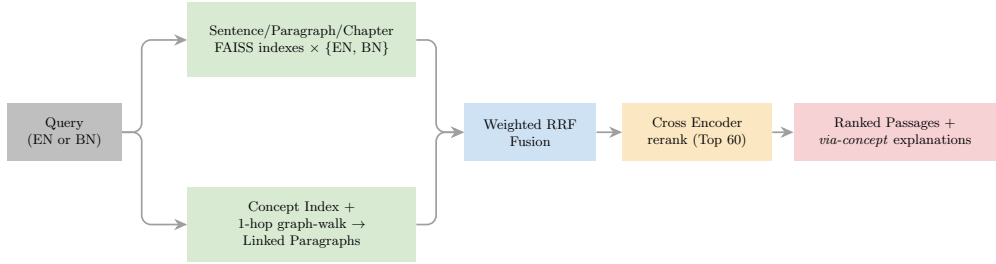


Fig. 1 Retrieval architecture. Path A (top) matches the query against three granularities per language; Path B (bottom) routes it through the concept graph. Both feed a single fusion and reranking stage.

Section 4.2’s stated budget, and are provided for convenience rather than as the primary artefact.

The underlying texts require a distinction. Vivekananda died in 1902 and his own words are in the public domain, but the modern published editions from which we parse — and, in particular, the Bengali renderings of English lectures, which are the work of later translators — carry editorial and translational rights held by their publishers. We therefore do *not* redistribute the source HTML. What we release are the derived layers plus the anchors needed to align them against the publicly hosted editions, so the resource is fully reproducible by a user who obtains the source texts through the publishers’ own channels, and the pipeline regenerates every layer from them. **[Confirm with the publishers of the English and Bengali editions before submission; if redistribution of the parsed text is permitted, release it and revise this paragraph.]**

6 Multi-Granular Fused Retrieval

All text units — 168k sentences, 33k paragraphs, 2k chapters, and the 8.4k concept labels — are embedded into the same 1024-dimensional space with BGE-M3 [5] and stored in exact inner-product FAISS indexes [7] (about 1.1 GB in total including meta-data, small enough that approximate indexing is unnecessary). Retrieval runs two paths per language (Figure 1):

Path A (direct). The query is matched against the sentence (top-30), paragraph (top-20), and chapter (top-10) indexes. Sentence and chapter hits are mapped to their enclosing/leading paragraphs, since the paragraph is the unit surfaced to users; the granularity that found each hit is retained and displayed.

Path B (concept-mediated). The query is matched against the concept-label index (top-12 anchors), the anchors are expanded one hop along *similar* and *co-occurs* edges with weight decay, and each activated concept contributes its top linked paragraphs. Because concept nodes are language-agnostic, an English query activates Bengali paragraphs (and vice versa) with no translation step. Path B also yields an explanation — the concepts through which each paragraph was reached — which the interface displays as “via concepts” badges.

Candidates from the four rank lists (three granularities + concept path) are fused with weighted reciprocal rank fusion [8], with the concept path weighted slightly above the direct paths (1.2 vs. 1.0) and chapters below (0.7). The top 60 fused candidates are reranked by the BGE-reranker-v2-m3 cross-encoder, and the top k per language are returned with full provenance (volume, section, chapter, paragraph index, and a deep link to the chapter anchor in the hosted edition).

7 “Ask Vivekananda”: Bridged, Citation-Grounded QA

The QA layer turns retrieval into direct answers under two hard constraints: modern questions must reach era-appropriate material, and every generated claim must be verifiable. The pipeline has three stages.

Stage 1: analyze and bridge. The question’s 15 nearest concept labels are fetched from the concept index. A single LLM call receives the question *and this list* and returns (i) 2–4 reformulated “timeless” search queries free of modern-only vocabulary, (ii) up to 6 concept labels *copied from the offered list*, and (iii) a one-line *bridge note* explaining the modern-to-timeless mapping. Restricting concept choice to labels that verifiably exist in the graph (validated again against the database after generation) prevents the bridge itself from hallucinating; if JSON parsing fails, the system degrades to retrieval with the raw question. The question’s language is detected deterministically from its script, never delegated to the LLM.

Stage 2: retrieve. The original question and each reformulation are run through the full retrieval stack of Section 6; results are merged, deduplicated per (language, paragraph), and the pooled candidates are reranked once against the *original* question. The top 10 passages become the numbered sources.

Stage 3: generate and ground. A locally hosted Qwen2.5-7B-Instruct [9] (served via Transformers [11]; any OpenAI-compatible endpoint can be substituted by configuration) receives the numbered sources — location-labelled, language-tagged, and trimmed to a window centred on each passage’s best-matching sentence — and composes a scholarly third-person answer in the language of the question. The system prompt forbids claims not supported by the sources and requires a bracketed citation after every substantive statement. Post-processing then rewrites each citation marker $[n]$ into a hyperlink into the published online edition and *deletes* any marker whose index does not correspond to a retrieved source, so a fabricated citation cannot survive to the reader. The published editions carry a stable HTML anchor on every paragraph, so each citation deep-links to the *exact cited paragraph* (falling back to the chapter anchor where a paragraph anchor is absent) — the reader lands on the sentence in question, not merely the top of a chapter. The interface additionally shows the bridge note, the reformulated queries, the chosen concepts, and, for every source, the matched sentence and the full passage — the entire evidential chain is one click deep.

Table 5 Serving-time characteristics (single NVIDIA H200; the system also runs CPU-only with seconds-level search latency). Model download sizes excluded.

Search, with cross-encoder rerank (mean of 5 queries)	0.28 s
Search, fusion only	0.08 s
QA end-to-end (bridge + retrieve + generate)	6–9 s
Index footprint (7 FAISS indexes + SQLite metadata/graph)	1.1 GB
Full pipeline build (one GPU)	< 4 h
Incremental rebuild after a source edit	10–15 min

8 System Characteristics

The stack is deliberately conservative: exact FAISS search, SQLite for all metadata and graph storage, and a Streamlit interface. There are no external service dependencies at query time; the system runs entirely on-premises, which matters both for cost and for institutions that cannot ship religious-community data to third-party APIs. Table 5 reports measured characteristics. A test suite of 74 CPU-only tests covers the parsers, schema, extraction-output parsing, fusion, bridging fallbacks, prompt budgeting, citation linkification, and the concept-graph view.

9 Evaluation

We report three studies: two automatic ones that require no manual annotation, and a human study of the concept layer. All are reproducible with evaluation scripts shipped in the repository.¹

9.1 Known-item cross-lingual retrieval

The *Vani o Rachana* contains independent Bengali renderings of many English lectures, which yields chapter-level cross-lingual relevance judgments at near-zero annotation cost. We identified candidate pairs as mutual-best matches between the chapter-level embedding indexes (cosine ≥ 0.70 ; 285 candidates), then verified each candidate with the local LLM shown both chapters’ titles and opening paragraphs, retaining $N = 194$ verified rendered pairs. For each pair, the source chapter’s title and opening paragraph (truncated to 600 characters) is the query, retrieval is restricted to the *other* language, and the paired chapter is the sole gold item; a retrieved paragraph counts if it belongs to the gold chapter. We report Recall@10 and MRR over the top 10 returned paragraphs in both directions (Table 6).

One caveat: pair discovery uses the same embedder as Path A (albeit on whole-chapter vectors, while queries are titles plus opening paragraphs), so absolute Path-A numbers may be slightly optimistic; comparisons *across* configurations are unaffected.

Three observations. First, the shared multilingual embedding space alone is a strong cross-lingual known-item retriever: a query built from an English lecture finds

¹scripts/eval.crosslingual.py, scripts/eval.qa.integrity.py, and scripts/eval.concept.precision.py; the raw outputs quoted here, including the per-annotator judgment files, are included under docs/paper/eval/.

Table 6 Known-item cross-lingual retrieval over $N = 194$ LLM-verified rendered lecture pairs (EN $\rightarrow$ BN / BN $\rightarrow$ EN).

Configuration	Recall@10	MRR
Path A (direct dense)	0.88 / 0.82	0.74 / 0.65
Path B (concept-mediated)	0.14 / 0.12	0.07 / 0.04
Fused (weighted RRF)	0.88 / 0.82	0.67 / 0.62
Fused + cross-encoder rerank	0.86 / 0.86	0.67 / 0.70

Table 7 Citation-integrity and bridging statistics over 30 questions (20 EN / 10 BN; 15 modern / 15 timeless).

Questions answered (non-empty retrieval)	30 / 30
Stage-1 JSON fallback rate	0%
Bridge note produced: modern questions	14 / 15 (93%)
Bridge note produced: timeless questions	2 / 15 (13%)
Citation markers per answer sentence	0.47
Citation markers emitted (total)	136
Fabricated markers removed by validator	0 / 136 (0%)
Cross-lingual share of top-10 evidence (bridged)	32%
Cross-lingual share of top-10 evidence (raw question)	40%

its Bengali rendering among 14,455 Bengali paragraphs at Recall@10 of 0.88. Second, the concept path is *not* a known-item retriever and is not meant to be one — it expands a query into thematically related material across the whole graph, which is precisely what known-item metrics penalize; its value shows in query bridging and in the explanatory “via concepts” chains, not in this table. Third, fusion preserves Path A’s recall while adding the concept candidates, and the cross-encoder markedly improves the harder BN $\rightarrow$ EN direction (MRR 0.62 $\rightarrow$ 0.70, Recall@10 0.82 $\rightarrow$ 0.86) at a small cost in EN $\rightarrow$ BN, yielding the most balanced configuration — the production default.

9.2 Citation integrity and bridge reliability

We ran the full QA pipeline over a fixed set of 30 questions — 20 English and 10 Bengali; 15 phrased in deliberately modern vocabulary (smartphone addiction, social-media envy, job-loss anxiety, burnout) and 15 in timeless vocabulary (fear, concentration, duty, anger) — and logged the pipeline’s internal decisions (Table 7).

The bridging stage discriminates almost perfectly between question types: it produced a bridge note for 93% of modern questions and only 13% of timeless ones, with a 0% structured-output failure rate. Generated answers cite densely (one marker per two sentences on average), and in this run the 7B model emitted *no* citation marker pointing outside the retrieved source list — the validator, which deletes such markers, acted as an idle guardrail rather than an active repair mechanism.

The bridge ablation produced a finding we did not anticipate: the cross-lingual share of the evidence pool is essentially unchanged by bridging (32% bridged vs. 40%

raw; 47% vs. 45% on the modern subset). Cross-linguality is supplied by the shared embedding space and the concept graph themselves, not by the reformulation step. The bridge’s measured contributions are therefore selective activation (firing when and only when the vocabulary gap exists), the era-appropriate reformulations, and the user-facing explanation — not an increase in cross-lingual recall, which the underlying retrieval already provides.

9.3 Concept-extraction precision: a human study

The concept layer is the system’s central claim, and it is produced entirely by an LLM. We therefore ran a human study of its precision.

Protocol. From all 87,518 paragraph–concept edges we drew a uniform random sample of $N = 200$ (seed 13). Each item presents one paragraph, one attached concept label, and the extracted relation. Annotators judged each item as *correct* (the concept is a correct reading of the paragraph *at the stated relation*) or *incorrect* (wrong, merely tangential, or right concept but wrong relation). Judgments were collected through a small purpose-built web interface shipped with the repository, which saves after every click and lets an annotator resume; no annotator saw another’s judgments or the extractor’s confidence weight.

Annotators. Three annotators, all familiar with the corpus, each judged all 200 items independently. Annotators A and B are fluent in both English and Bengali; annotator C is proficient principally in Bengali. Since 85 of the 200 sampled items are English, we use C’s judgments only for the Bengali subsample, where that proficiency applies, and base the whole-sample figures on A and B. All three judgment files ship with the repository.²

That scoping is a competence requirement fixed by the task, and the measured reliability tracks it closely: against A, annotator C reaches $\kappa = 0.49$ on the Bengali items but only $\kappa = 0.14$ — barely above chance — on the English ones. Restricting C to Bengali is therefore not a way of setting aside an inconvenient judge but of using each annotator where they are reliable; we report the Bengali three-annotator panel in full below, and it does not flatter the system.

Disclosure. Annotator B is an author of this paper; annotators A and C are not. B is marginally the more lenient of A and B (0.705 vs 0.665), so we treat the strict consensus figure — not B’s — as the headline number.

Precision is moderate; agreement is substantial. Consensus precision is 0.60 — three in five sampled edges are endorsed by both judges — and Cohen’s κ of 0.61 falls in the “substantial” band on the conventional scale [15]. The residual disagreement is not systematic: 21 items go one way and 13 the other, which a McNemar exact test does not distinguish from chance ($p = 0.23$). The two are applying the same rubric with ordinary noise, which is what an inter-annotator agreement study is supposed to establish before its precision figure is worth quoting.

The extractor’s confidence weight is calibrated. Each paragraph–concept edge carries a weight the extraction LLM assigns itself. It predicts human judgment

²`docs/paper/eval/annotations/`; the whole-sample figures below are reproduced with `eval.concept.precision.py compute --users annotator_a annotator_b`, and any other pairing can be recomputed from the same files.

Table 8 Concept-extraction precision over $N = 200$ randomly sampled paragraph–concept edges, annotators A and B. Intervals are Wilson 95% (bootstrap for κ).

Measure	Value	95% CI
Precision, annotator A (non-author)	0.665	[0.597, 0.727]
Precision, annotator B (author)	0.705	[0.638, 0.764]
Precision, consensus (both correct)	0.600	[0.531, 0.665]
Precision, union (either correct)	0.770	[0.707, 0.823]
Raw agreement	0.830	—
Cohen’s κ [14]	0.607	[0.482, 0.720]
Gwet’s AC_1 [16]	0.701	—

Table 9 Precision by extractor confidence weight. The high-confidence subgraph is 35.9% of all edges but still covers 98.1% of the paragraphs that carry any concept.

	Annot. A	Annot. B	Consensus
weight ≥ 0.8 ($n = 69$)	0.783	0.812	0.710
weight < 0.8 ($n = 131$)	0.603	0.649	0.542

(Spearman $\rho = 0.19$ against consensus), and thresholding on it works (Table 9): consensus precision rises from 0.54 below 0.8 to 0.71 at 0.8 and above (Fisher exact $p = 0.023$).

The pruning is nearly free in coverage terms. Applied to the full graph, weight ≥ 0.8 retains 31,445 of 87,518 paragraph–concept edges (35.9%) and 3,709 of 8,362 concepts (44.4%) — but 22,914 of the 23,363 paragraphs that carry any concept (**98.1%**) keep at least one. The low-confidence mass is redundant attachment to already-covered paragraphs, not the sole handle on distinct material. A deployment that needs the graph to be *right* rather than *broad* should therefore threshold at 0.8; we report the unpruned graph throughout this paper because Path B is used for recall-oriented expansion, where the cost of a wrong concept is a wasted glance rather than a false statement.

Extraction is weaker in Bengali. Consensus precision over A and B is 0.68 on English items and 0.54 on Bengali (Fisher exact $p = 0.043$). The Bengali subsample ($n = 115$) is the one part of the study carrying a third independent judgment, and the fuller panel confirms the gap rather than closing it: across A, B and C the three-rater Fleiss κ is 0.49 and majority-of-three precision is 0.59, still well below the 0.68 measured on English. Adding a Bengali-proficient annotator thus corroborates the finding with the language’s strongest available judgement.

This is the study’s one clearly actionable quality finding beyond the threshold: the extraction LLM is a predominantly English-trained model reading Bengali prose, and the concept layer inherits that asymmetry. Since cross-lingual bridging is the system’s central claim, the weaker half of the graph is precisely the half doing the work of

making Bengali material reachable from English queries — a Bengali-strong extractor, or a second extraction pass over Bengali paragraphs alone, is the most valuable single upgrade available to the pipeline.

Relations. The dominant *discusses* (87.5% of the sample) reaches consensus precision 0.62, while the two rare relations are worse (*exemplifies* 0.47, $n = 17$; *contrasts* 0.50, $n = 8$) — on samples too small to support a strong claim, but consistent with the intuition that a directional judgment is harder for the extractor than a topical one.

On the rubric. The binary judgment conflates two readings of “the concept describes the paragraph”: whether the paragraph *is about* the concept, and whether the concept is merely *present* in it. For a Sūtra gloss such as “The success of Yogis differs according as the means they adopt are mild, medium, or intense” tagged **success**, or a Bengali travel note announcing a forthcoming Vedanta lecture tagged **vedanta**, the first reading rejects and the second accepts. Both annotators converged on the more permissive presence reading, so the 0.60 we report should be understood as a precision under that reading; a study that fixed the stricter aboutness criterion for every annotator would return a lower number. A revised study should use a three-point scale (*central* / *mentioned* / *unsupported*) rather than force this distinction into a binary, and should fix the reading in the guidelines instead of leaving it to be resolved per annotator.

What the errors look like. The characteristic failure is not a bizarre concept but a plausible one attached to a paragraph that does not support it. Narrative, epistolary, and connective passages attract the corpus’s thematic vocabulary regardless of content: a Bengali letter remarking that a book “has been well received here, but the publishers do not seem to be pushing sales” (our translation) is tagged **knowledge-realization**; a subscription appeal for a memorial is tagged **liberation**. The extractor appears to condition on the corpus’s overall register as much as on the paragraph in front of it. This is precisely the failure mode that the confidence threshold suppresses.

Implication for the system’s claims. The concept layer should be understood as a *recall-oriented, imperfect expansion and explanation layer*, not a curated ontology. This is consistent with Table 6, where Path B is a poor known-item retriever on its own; its contribution is cross-lingual reach and the “via concepts” explanations, both of which degrade gracefully under error because every retrieved passage is displayed to the reader with its source. A wrong concept costs an irrelevant result, not a false assertion about Vivekananda — the citation discipline of Section 7, not the concept graph, is what bounds that risk.

10 Case Studies

We illustrate the bridging behaviour with unedited system outputs.

Modern question, English. For “*How do I get rid of mobile phone addiction?*” the analysis stage produced the bridge note “*Mobile-phone addiction is a modern form of attachment of the senses and a habit that enslaves the mind*”, selected the corpus concepts **excessive-attachment** and **self-control**, and reformulated the question into queries such as “*How can I overcome excessive attachment to sense objects?*” and “*What is the remedy for habit that enslaves the mind?*”. Retrieval returned ten

passages spanning *Inspired Talks*, the Raja-Yoga lectures, and — notably — Bengali chapters on liberation (*mukti*) and sense-withdrawal (*pratyahara*), which no lexical and no unbridged dense query for “mobile phone” reaches. The generated answer opens by restating the bridge, then prescribes non-attachment and graded practice of sense-withdrawal, with every paragraph citing its sources.

Modern question, second example. For “*How should one deal with social media envy?*” the bridge note read “*Social media envy is a modern form of coveting the social status and approval of others. . .*” — a mapping to covetousness and approval-seeking that the authors did not anticipate but that the concept inventory supported.

Bengali question. A Bengali question on freedom from mobile addiction (“*mobile asakti theke muktir upay ki?*”) was detected as Bengali, bridged through the same English-labelled concepts, answered *in Bengali*, and grounded predominantly in Bengali sources while still drawing on English lectures — the concept layer making the cross-lingual evidence pool available in both directions.

11 Limitations and Ethical Considerations

An imperfect concept layer. The human study of Section 9.3 reports strict-consensus precision of 0.60 over paragraph–concept edges, and 0.54 on Bengali material. The concept graph is a useful expansion and explanation layer, but it is not a scholarly index and should not be cited as one. Users who need higher precision should threshold at extractor weight ≥ 0.8 , which lifts consensus precision to 0.71 at almost no cost in paragraph coverage.

Limits of that study. The whole-sample figure rests on two annotators, one of them an author of this paper and marginally the more lenient of the two, judging 200 items — enough to place the precision within roughly ± 0.07 , but not to support fine comparisons. A third annotator, proficient principally in Bengali, contributes only to the Bengali subsample; the English half therefore carries no judgement from outside the author’s immediate circle. The binary rubric conflated “the paragraph is about this concept” with “this concept appears in the paragraph” and left the choice to each annotator; the two whole-sample annotators settled on the more permissive reading, so the headline figure is the optimistic end of the plausible range — the Bengali-proficient annotator, who judged more strictly, would place it lower. A three-point scale, guidelines that fix the reading, and a larger and wholly independent annotator pool are all needed before the precision figure can be treated as settled rather than indicative.

Answer faithfulness is still unevaluated. Citation integrity is measured structurally (marker validity), not semantically: we verify that every citation points at a retrieved passage, not that the passage supports the claim attached to it. Known-item judgments likewise come from rendered lecture pairs rather than topical relevance assessment. A scholar-annotated faithfulness study of generated answers remains the most important piece of future work.

Generation quality floor. The default 7B answer model occasionally paraphrases loosely when quoting, and may present material from a Bengali source in English without consistently marking it as translated, despite prompt instructions.

The citation mechanics bound the damage — the linked passage is always one click away — but do not eliminate it. Configurable substitution of a stronger model raises the floor.

Extraction bias. The concept inventory reflects the judgement of the extraction LLM; systematic blind spots in that model’s reading of Vedantic material would propagate to the graph. The alias tables and per-paragraph provenance make audits feasible.

Cultural sensitivity. The corpus is scripture-adjacent for a living community. We consider verifiable grounding, the visible bridge explanation, and on-premises deployment to be ethical requirements for this material, not merely engineering choices, and we would caution against deploying fluent-generation systems over such corpora without citation enforcement of comparable strictness.

12 Conclusion

VIVEKA-INSIGHT demonstrates that a modest, fully local pipeline — one extraction pass by a mid-sized LLM, a shared multilingual embedding space, and a canonical-label concept graph — suffices to make a bilingual, non-parallel classical corpus searchable by meaning, answerable in either language, and honest about its evidence. The temporal bridge turns the system’s most obvious weakness (a nineteenth-century vocabulary) into an explicit, inspectable mapping rather than a silent retrieval failure. The architecture assumes nothing Vivekananda-specific: any corpus with stable canonical structure and one or more languages covered by the embedding model — the Ramakrishna *Kathamrita* literature, Gandhi’s collected works in English, Gujarati and Hindi, or patristic corpora in Greek and translation — could be indexed with the same pipeline. The human study also marks the clearest directions for the next iteration. At strict-consensus precision of 0.60 the concept layer is broad but imperfect, and two of its properties point at cheap gains: the extractor’s own confidence weight separates good edges from bad ones well enough that a calibrated threshold buys accuracy almost for free, and the precision gap between English (0.68) and Bengali (0.54) says the largest single quality deficit is not the pipeline but the extraction model’s weaker reading of Bengali — the very half of the corpus that cross-lingual bridging exists to open up. A Bengali-capable extractor and community-driven correction through the audit trail the system already exposes are therefore the first steps; graph-embedding refinement of the concept layer and a scholar-annotated faithfulness study of generated answers are the further ones.

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Competing interests. The author declares no competing financial interests. One non-financial interest is disclosed: the author served as one of the three annotators in the concept-precision study of Section 9.3. The study reports each annotator separately, treats the strict two-annotator consensus rather than the author’s own more lenient judgments as the headline figure, and releases all per-annotator judgments so that any aggregation can be recomputed independently.

Ethics approval. Not applicable. The study involved no human subjects: the three annotators contributed expert judgments on published text as acknowledged contributors, not as research participants, and no personal data were collected.

Data availability. The derived resource layers described in Section 5 — structural parse, concept graph, alias inventory, entity layer, and the 200-item annotated evaluation set with all per-annotator judgments — are archived under CC BY 4.0 at 10.5281/zenodo.21669974. The source texts of the published editions are not redistributed; see Section 5.3.

Code availability. The complete construction and serving pipeline, including the three evaluation scripts that reproduce every figure reported here, is released under the MIT licence at <https://github.com/tamalrkm/viveka-insight>.

Author contributions. T.M. designed and implemented the system, conducted the evaluations, and wrote the manuscript.

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